

Customer: Tong Hsing Electronics Taiwan **Date:** 19-Jan-17 **Time:** 09:30

Participant From Maruwa	Participant From Organizer	Report To
1. Mr Sasaki MJ R&D	1. Mr Brian Prod Mgr	1. Mr.Anthony,Mr.Ogawa
2. Mr Siva MMSB QA	2. Ms Polly Purchasing	2. Mr.Seeni,Mr.Naga,Mr.Sinna
3. Mr Vega Maruwa Taiwan	3. Mr Mark QA	3. Mr.Theva,Mr.Shahrudin,Mr.Fauzi
4 Mr Eric	4 Mr Klinsmen QA Manager	4 Mr. Matsumoto
5.	5.	5.

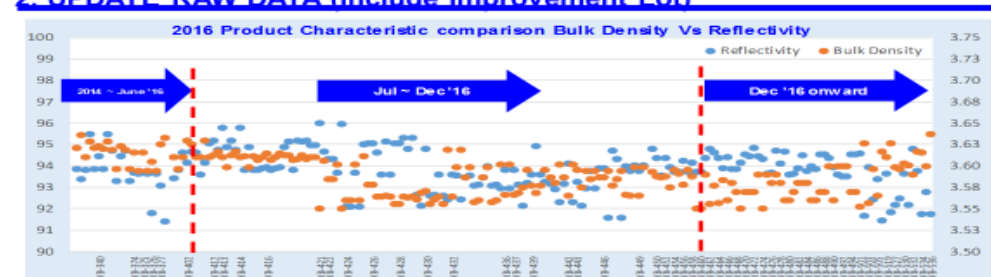
Agenda

Meeting starts at 930am

Quality Disucssion

Meeting started with presentation from Maruwa. All the previous finding have been explained and a few question raised by TH members.Below is the questions

2. UPDATE RAW DATA (Include Improvement Lot)



- 1) ~ May 2016 : Specification (Based on QDS)
 - Reflectivity : 91% ~ 97%
 - Specific Gravity : Above 3.50
 - Water absorption : Max 5.0%
- 2) June 2016 ~ Nov 2016 : Request by Tong Hsing during visit to Maruwa on June 2016
 - Reflectivity : 91% ~ 97%
 - Specific Gravity : Above 3.5 ~ 3.7
 - Water absorption : Max 5.0%
- 3) Dec 2016 onward : Revise specification
 - Reflectivity : 91% ~ 94%
 - Specific Gravity : Min 3.55
 - Water absorption : Max 1.0%

Improvement LOT
WB-501

TONG HSING: Required Maruwa to explain why 2014 showing smaller range both in SG & Reflectivity but range increase in between.

MARUWA: Explain again to TH that MARUWA focus only for the reflectivity, Bulk density & Water absorption specs. No point of reviewing the past data since it is already happen and measurement still within the QDS spec. Focus on the DEC16 onwards lot

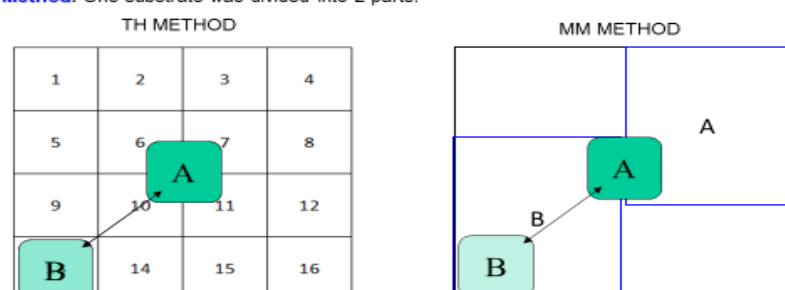
TONG HSING: Why is the centre value and edge value is different. Even TH measured the centre they are getting the value low.

MARUWA: Explain again the theory of ceramic when fire. Maruwa also have standardize the measurement by taking edge measurement. Request to move forward by standardize the measurement method which to be carry out today.

5.0 BULK DENSITY MEASUREMENT VARIENCE (MW & TH)

Result of Bulk Density & Water absorption

TH Method: One substrate was divided into 16 parts.
MM Method: One substrate was divided into 2 parts.



Question: TH 16pcs data & Maruwa data shows a difference of 0.1g/cm3. TH would like to know the reason of this difference.
Ans : The Bulk Density value at center is higher than the edge. MM method measure include center and edge area, but TH method measure by individual area. That's the reason why its has 0.1g/cm3 variation.

MARUWA MASTER SAMPLE MEASUREMENT

BULK DENSITY

SAMPLE	MARUWA	TONG HSING	VAR	%VAR
A	3.675	3.63	0.045	1.22%
B	3.665	3.64	0.025	0.68%
C	3.675	3.59	0.085	2.31%
		AVERAGE	0.052	

There is difference between Maruwa & TH by 0.052g/cm3

The difference is cause by a few abnormal at the measurement method.

COMPARISON OF MEASUREMENT METHOD MARUWA & TONG HSING

AREA	MARUWA	TONG HSING
METHOD	ULTRASONIC (60min)	ULTRASONIC (60min)
	WEIGHT IN WATER	WIPE & WET WEIGHT
	WIPE & WET WEIGHT	WEIGHT IN WATER
	OVEN	OVEN
	DRY WEIGHT	DRY WEIGHT
MACHINE	4 DECIMAL POINT	UNIT 1 : 2 decimal UNIT 2 : 3 decimal

2 major problem detected during observation is

- a. Did not follow the standard flow
- b. Measurement equipment showing smaller decimal

Impact on decimal is

EG: TH equipment => 1.20gm

Chances: 1.195~1.204

When calculate back, the data shows variance

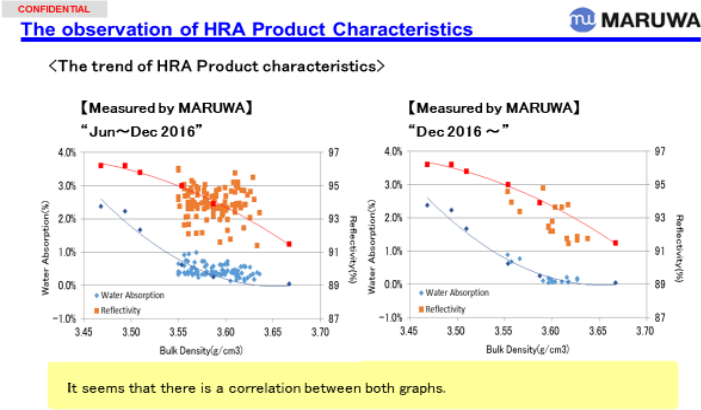
AVG: 0.080 MAX: 0.144 MIN: 0.046

ACTION by Tong Hsing : To combine the 4 corner sample and measure as one (temporary action

MARUWA: Any abnormal due to this is not MARUWAs reponsibility. MARUWA will remain the current method and confirm all the 4 corners

(This is a temporary action by TH until they purchase a new unit with 3 decimal point)

TONG HSING INCOMING MEASUREMENT DATA ANALYSIS



TH measurement doesn't shows the value falls on HRA standard. TH data showing difference especially Bulk density versus reflectivity. Sample taken to be remeasure
PIC: MR SASAKI

MARUWA (MALAYSIA) SDN. BHD

Meeting adjourned at 4:30pm

Conclusion & Opinion

All the pending issues been closed and informed to TH top management to focus on the future incoming. There is a lot finding especially on the measrement method and all problems have been highlighted to them. Currently TH request Maruwa to produce for 3.6g/cm3. There is some request from TH to Maruwa that to run 2 firing machine for density 3.55g/cm3 & 3.6g/cm3. Informed to TH that this topic to be dsicuss with Taiwan Sales & MW PC. We requested TH to follow the flow that been explain earlier to avoid any wrong info. Next action to concentrate n the 3.6g/cm3 and 3.65g/cm3 (HRA TYPE C)

No	Complaint / Request	Image	ACTIVITY	PIC	DATE																	
1	SG measurement variance	NA	To confirm the string height and impact for the density value	MrAmni																		
2	String control		Currently control by calibration division	Ms Watie																		
3	Density Trend Data		To review the trend data from 2014 why the trend fluctuate	Mr Siva																		
5	Measurement method	<table><tr><td>AREA</td><td>MARUWA</td><td>TONG HSING</td></tr><tr><td rowspan="5">METHOD</td><td>ULTRASONIC (60min)</td><td>ULTRASONIC (60min)</td></tr><tr><td>WEIGHT IN WATER</td><td>WIPE & WET WEIGHT</td></tr><tr><td>WIPE & WET WEIGHT</td><td>WEIGHT IN WATER</td></tr><tr><td>OVEN</td><td>OVEN</td></tr><tr><td>DRY WEIGHT</td><td>DRY WEIGHT</td></tr><tr><td>MACHINE</td><td>4 DECIMAL POINT</td><td>UNIT 1 : 2 decimal UNIT 2 : 3 decimal</td></tr></table>	AREA	MARUWA	TONG HSING	METHOD	ULTRASONIC (60min)	ULTRASONIC (60min)	WEIGHT IN WATER	WIPE & WET WEIGHT	WIPE & WET WEIGHT	WEIGHT IN WATER	OVEN	OVEN	DRY WEIGHT	DRY WEIGHT	MACHINE	4 DECIMAL POINT	UNIT 1 : 2 decimal UNIT 2 : 3 decimal	2 major problem detected during observation is a. Did not follow the standard flow b. Measurement equipment showing smaller decimal Impact on decimal is EG: TH equipment ==> 1.20gm Chances: 1.195~1.204 When calculate back, the data shows variance AVG: 0.080 MAX: 0.144 MIN: 0.046	Mr Siva	
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6	Sample confirmation		3 sets of sample provided to confirm bulk density. TH confirm is out of spec. Measured by TH is below 3.55g/cm3. Maruwa will re-confirm the data	Mr Amni/Mr Hamdan																		
7	WARP		Detected LOT NO: WBB-527 showing WARP upper trend and require MARUWA to confirm it. Product is still within the spec	Mr Siva																		
8	Delivery		CREE & TH would like to know the current input material will be shipout when? This is for them to understand when the transition period can take place for 3.6g/cm3	Mr Theva	24/01/2017																	